

## PHASE 7

# Project Documentation Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

| Project Focus                                                                            |
|------------------------------------------------------------------------------------------|
| AI-assisted learning resources, secure backend services, and personalized study support. |

## Phase Objective

To consolidate the design, implementation, configuration, testing, and usage information into a clear technical record that can be maintained and presented.

## 7.1 Documentation Coverage

- Project description and problem background.
- User problems and proposed solution.
- Functional and technical requirements.
- Software and hardware requirements.
- System architecture and component responsibilities.
- Data entities and relationships.
- MVC organization.
- Project setup and environment configuration.
- Backend modules and their responsibilities.
- Database configuration and schemas.
- API endpoints.
- Execution and testing procedure.
- Demonstration references.

## 7.2 Technical Records

- Document the Node.js and Express.js backend structure.
- Record MongoDB and Mongoose usage.
- Describe JWT and role-based security.
- Explain Gemini integration and the supported AI operations.
- Maintain the endpoint list used for API testing.
- Keep configuration requirements documented without exposing secret values.

## 7.3 User-Oriented Documentation

The documentation should explain the normal student journey from registration through material upload and AI-resource generation, while also describing administrator responsibilities for account and system management.

## 7.4 Maintenance Value

A well-organized record helps future developers understand how the modules interact, locate configuration requirements, reproduce the local setup, and extend the platform without redesigning the existing architecture.

## 7.5 Supporting Evidence

- Backend execution/demo reference: AI\_Studybuddy\_Backend\_Demo.mkv.
- API testing/demo reference: AI\_Studybuddy\_Execution\_Demo.mkv.
- The source document also contains a Google Drive demonstration reference and a code-drive reference.

## 7.6 Documentation Result

The completed project record acts as a single technical reference for understanding the purpose, structure, configuration, implementation, testing process, and demonstration of AI StudyBuddy.